

Generative model and core theory for `mcpopgen`

Sequence $\rightarrow$ TF occupancy $\rightarrow$ methylation $\rightarrow$ (mutation, fitness) in a Wright–Fisher population

working notes

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This document states exactly what the simulations compute. Every quantity here has a counterpart in `scripts/slim/*.slim` (SLiM 5.2) and `scripts/python/model.py`. Section 3 gives the per-generation algorithm, Sections 4–6 the maps, Section 7 the model set, Section 8 the results used to validate the simulator (experiment E0), and Section 9 the predictions for experiment E1.

1 What the experiments are

E0 (calibration). No methylation or TF. It checks that the SLiM machinery (mutation matrix, rejection gating, selection, rescaling) reproduces quantities that can be derived exactly: neutral diversity, the site-frequency spectrum, Kimura fixation probabilities, and CpG decay. It says nothing biological. It certifies the simulator.

E1 (passenger footprint). Methylation is computed from sequence and can gate CpG mutation, but it *never* enters fitness. Question: how much CpG enrichment near a selected TF motif arises with zero selection on any CpG?

E2 (causal vs passenger; next). Same maps; fitness reads methylation (model C) or occupancy (model P). Question: which observable statistics separate them once mean methylation is matched?

2 State space and notation

- A locus of L bases (E1: $L = 300$). A haplotype is $x = (x_0, \dots, x_{L-1}) \in \{A, C, G, T\}^L$, coded 0, 1, 2, 3.
- The population has N diploid hermaphrodites, i.e. $2N$ haplotypes. Individual j is a pair $(x^{(j,1)}, x^{(j,2)})$. There is no recombination within the locus.
- CpG set: $\mathcal{C}(x) = \{i : x_i = C, x_{i+1} = G\}$, indexed by the position i of the C.
- TF motif: a fixed 8-mer $\mu = \text{TGACTCAT}$ (it contains no CpG), at positions $M = \{s, \dots, s+7\}$, with $s = L/2 - 4$ and centre $c = s + 3.5$.
- Distance of position i from the motif edge: $d_i = \max(0, |i - c| - 4)$.
- $\sigma(z) = 1/(1 + e^{-z})$.

Methylation is **not** a state variable. It is recomputed from each haplotype’s sequence whenever it is needed. This encodes germline reprogramming: there is no transgenerational epigenetic inheritance, so methylation is a deterministic molecular phenotype of the haplotype (in cis), averaged over cells.

3 One generation (SLiM’s Wright–Fisher cycle)

Let generation t consist of N individuals with fitnesses $w_1, \dots, w_N$ (computed at the end of generation $t - 1$).

1. **Parent choice.** Each of the N offspring draws two parents independently, each with probability $w_j / \sum_k w_k$. Incidental selfing (the same parent drawn twice, probability $\approx 1/N$) is allowed, as in the standard WF model.
2. **Gamete formation.** Each parent transmits one of its two haplotypes, chosen with probability $1/2$. There is no recombination.
3. **Mutation.** The transmitted haplotype x receives new mutations by the context-dependent process of Section 4, including the methylation gate (evaluated on x , the parental sequence).
4. **Bookkeeping and sampling.** `late()` events record statistics. Fitness w of each new individual is then computed from its two haplotypes (Section 6) and used in step 1 of the next generation.

Selection is therefore fecundity/viability selection applied once per generation, and drift is multinomial sampling of $2N$ haplotypes. The effective size is $N_e = N$ (diploid WF), which is the convention for every $2Ns$ and $4Nu$ below.

4 Mutation process

4.1 Context-only (“ctx”) process

Each haplotype base i mutates to base $b \neq x_i$ at rate (per generation)

$$q_i(b \mid x) = \begin{cases} r\alpha & \text{if } (x_i, x_{i+1}, b) = (C, G, T) \text{ or } (x_{i-1}, x_i, b) = (C, G, A) \\ \alpha & \text{otherwise.} \end{cases} \quad (1)$$

This is Jukes–Cantor (total $u = 3\alpha$ per site) plus CpG transition hypermutability r . In SLiM it is a 64×4 trinucleotide matrix (rows are (x_{i-1}, x_i, x_{i+1}) , columns are b). Bases off either end count as A (SLiM’s convention), so position 0 is never the G of a CpG.

4.2 Methylation-gated (“meth”) process

Deamination of 5mC is what makes CpGs hypermutable, so only the *methyated* fraction carries the excess rate:

$$q_i^{\text{meth}}(T \mid x) = \alpha [1 + (r - 1) m_i(x)] \quad \text{for } i \in \mathcal{C}(x), \quad (2)$$

and likewise for the G (G→A) of the same CpG, using the same m_i . All other rates are as in (1). An unmethylated CpG ($m = 0$) has the ordinary transition rate α , and a fully methylated one has $r\alpha$.

Implementation by thinning. SLiM proposes CpG transitions at the maximal rate $r\alpha$ (from the matrix). A `mutation()` callback then accepts each proposal with probability

$$g(m) = \frac{1 + (r - 1)m}{r} \in [1/r, 1]. \quad (3)$$

Lemma 1 (thinning). *If events occur as a Poisson process of rate λ and each is kept independently with probability g , the kept events form a Poisson process of rate $g\lambda$. Hence accepted CpG transitions occur at exactly rate (2).*

SLiM already uses thinning internally: it proposes at the matrix’s maximal row rate and rejects according to the actual context. The gate is a second, state-dependent thinning. One caveat matters. SLiM passes the *parental* haplotype to the callback, so m_i is evaluated on the sequence being copied. That is exact only without recombination, which is why the methylation models have none. E0 verified that a constant gate $g = 0.5$ at $r = 20$ is indistinguishable from ctx at $r = 10$.

5 Maps: sequence $\rightarrow$ occupancy $\rightarrow$ methylation

TF occupancy (per haplotype). With $S(x) = \sum_{k=0}^7 \mathbf{1}[x_{s+k} = \mu_k]$ the number of matches,

$$O(x) = \sigma(\beta(S(x) - S_0)), \quad \beta = 2, S_0 = 6. \quad (4)$$

This gives $O = 0.982, 0.881, 0.5, 0.119$ for $S = 8, 7, 6, 5$. It is the simplest stand-in for a PWM/binding-energy model: occupancy saturates for good sites and collapses after two mismatches.

Methylation of the CpG at $i \in \mathcal{C}(x)$:

$$m_i(x) = \sigma(a_0 - a_1 \rho_i(x) - a_2 O(x) e^{-d_i/\lambda}), \quad \rho_i(x) = \#\{j \in \mathcal{C}(x) \setminus \{i\} : |j - i| \leq W\}. \quad (5)$$

- $a_0 = 3$ is the default state: $m = \sigma(3) = 0.953$ (bulk mammalian CpGs are mostly methylated).
- $a_2 = 6$ and $\lambda = 25$ set TF protection, i.e. low-methylated regions around bound sites (Stadler et al. 2011). A CpG next to a bound motif has $m = \sigma(3 - 6 \cdot 0.982) = 0.053$.
- a_1 and W set CpG-density feedback: dense CpG clusters resist methylation (CGI-like). With $a_1 = 0$ the map is local and per-CpG. With $a_1 > 0$ it is *degenerate*: many sequences give the same m , and selection on m spreads over many sites.

A Python mirror (`model.py`) recomputes m_i from SLiM’s output sequences. Agreement is $< 10^{-6}$, the limit of printed precision.

6 Expression and fitness

Diploid expression is additive (codominant) across the two haplotypes, and fitness is stabilising around the optimum $E^* = 1$:

$$E_j = \frac{1}{2} [g(x^{(j,1)}) + g(x^{(j,2)})], \quad w_j = \exp(-\kappa(1 - E_j)^2). \quad (6)$$

The haplotype readout $g(\cdot)$ is what distinguishes the models:

$$\text{P (passenger): } g(x) = O(x); \quad \text{C (causal, E2): } g(x) = 1 - \bar{m}_{\text{prom}}(x); \quad \text{M0 (neutral): } \kappa = 0. \quad (7)$$

Optimum. E1 uses $E^* = 1$. Because $O \leq 0.982$, that is directional selection with diminishing returns, not true stabilising selection. E2 uses an intermediate optimum (e.g. $E^* = 0.5$). Then many genotypes sit at the optimum, which allows compensatory turnover and lets methylation be conserved while sequence diverges.

Reporting selection. κ by itself is not interpretable. For a genotype G we report $2Ns_G$, with $s_G = 1 - w_G/w_{8/8}$. κ is set so that one heterozygous motif mismatch has $2Ns_{8/7} = 10$. That gives $\kappa = 11.7$ at $N = 100$ and $2Ns = 30, 108, 181$ for $7/7, 8/6, 8/5$. The motif is under strong selection and CpGs are under none.

7 The model set and the causal/passenger question

model	fitness reads	mutation	CpG feedback	role
M0_ctx	nothing	ctx	–	neutral baseline (E0’s process)
M0_meth	nothing	meth	$a_1 = 0$	motif decays, so no protection
M0_meth_fb	nothing	meth	$a_1 = 0.5$	can CpG islands self-sustain?
P_ctx	O	ctx	–	selection on TF, no gating
P_meth	O	meth	$a_1 = 0$	passenger footprint
P_meth_fb	O	meth	$a_1 = 0.5$	passenger + feedback
C (E2)	$\bar{m}$	meth	swept	methylation causal

Remark 1 (why P vs C is a statement about the genotype–fitness map). Because m is a function of x , every model is “selection on sequence”. P and C share both maps ($x \mapsto O$, $x \mapsto m$) and differ only in which composition, $w \circ O$ or $w \circ m$, defines fitness. A mutation $x \rightarrow x'$ can discriminate them only if it changes one readout but not the other. Examples: a CpG gain or loss outside the motif changes m but not O ; a motif mismatch with buffered methylation changes O but barely changes m . Such a mutation has a detectable fitness effect under C only if its selection is not swamped by drift, i.e. $2N |s(x \rightarrow x')| \gtrsim 1$. For a per-CpG effect Δm , $s \approx 2\kappa(1 - E) \Delta m/2$ (to first order), so identifiability requires

$$2N\kappa(1 - E) |\Delta \bar{m}| \gtrsim 1. \quad (8)$$

Two consequences follow. Under a degenerate map ($a_1 > 0$, near saturation), single-CpG $|\Delta \bar{m}|$ is small, so individual sites look neutral even when $\bar{m}$ is strongly conserved. And methylation-gated mutation (E1) can create CpG patterns that look like constraint with no selection at all.

8 Theory used for validation (E0)

8.1 Fixation probability

With genotype fitnesses $1, 1 + s/2, 1 + s$, the diffusion has drift $M(p) = \frac{s}{2}p(1 - p)$ and variance $V(p) = p(1 - p)/2N$. The scale function gives $u(p) = \frac{1 - e^{-2Nsp}}{1 - e^{-2Ns}}$, so a single new copy ($p = 1/2N$) fixes with

$$u = \frac{1 - e^{-s}}{1 - e^{-2Ns}} \xrightarrow{s \rightarrow 0} \frac{1}{2N}. \quad (9)$$

E0: 20,000 trials at each of 8 settings, all within 1.3 SE.

8.2 Neutral diversity at finite sites

Two lineages coalesce after $T \sim \text{Exp}(1/2N)$ generations, so they are separated by total branch length $2T$. Under JC, the probability that a site is identical after branch length ℓ is $\frac{1}{4} + \frac{3}{4}e^{-4u\ell/3}$. Then

$$\pi = 1 - \mathbb{E} \left[\frac{1}{4} + \frac{3}{4}e^{-8uT/3} \right] = \frac{\theta}{1 + 4\theta/3}, \quad \theta = 4Nu. \quad (10)$$

With $\theta = 0.01$, $\pi = 0.009868$. E0 matched this at $N = 500, 250, 100$.

8.3 Neutral SFS

In a sample of n , $\mathbb{E}[\xi_i] = \theta/i$. Folding gives the shape $\propto 1/i + 1/(n-i)$. E0 obtained χ^2 of 24–33 on 24 degrees of freedom.

8.4 The single-lineage oracle

Lemma 2. *Assume no selection and a mutation process in which each haplotype’s rates depend only on its own sequence (true for `ctx` and `meth`, since methylation is *cis*). Then, for any haplotype sampled at generation t , $\mathbb{P}(x(t) = y) = [e^{tQ}]_{x(0),y}$, where Q is the generator of the per-sequence mutation chain on $\{A, C, G, T\}^L$.*

Sketch. Trace the sampled haplotype’s ancestry back to generation 0. Under neutrality, the genealogy (which parent and which haplotype was copied) is independent of the sequences. Along the ancestral path, the sequence changes only by mutation, with rates that depend only on the current sequence on that path. So $x(t)$ is the chain run for t steps from $x(0)$. Discrete versus continuous time differs at $O(\text{rate}^2)$. $\square$

Hence $\mathbb{E}[\#\text{CpG}](t)$ in the population equals that of a Gillespie simulation of one sequence. This is the `oracle.py` check, and it needs no equilibrium. Under selection the lemma fails, because genealogy and sequence become coupled. For P models the oracle instead freezes the motif, and SLiM minus oracle then measures the effect of motif polymorphism.

8.5 Rescaling

The diffusion limit depends on parameters only through Nu , Ns , Nr_{rec} and time in units of N . Running $(N/Q, uQ, sQ, t/Q)$ preserves these and costs Q^2 less. This is valid while per-generation rates stay small ($r\alpha Q \ll 1$, $sQ \ll 1$). E0 checked neutral statistics at $Q = 1, 2, 5$. E1 checks `P_meth` at $Q = 1, 2, 4$.

9 E1 prediction: a two-state approximation

Treat one dinucleotide position as either CpG or not. A CpG with methylation m is lost at rate

$$\ell(m) = \underbrace{2\alpha[1 + (r-1)m]}_{\text{C} \rightarrow \text{T}, \text{G} \rightarrow \text{A}} + \underbrace{4\alpha}_{\text{other changes of C or G}}. \quad (11)$$

A non-CpG becomes a CpG at rate $\gamma\alpha$, where $\gamma \approx 6/15$: 6 of the 15 non-CpG dinucleotides are one step from CG (CA, CC, CT, AG, GG, TG), if dinucleotides were uniform. The equilibrium CpG frequency per position is then

$$f(m) \approx \frac{\gamma\alpha}{\gamma\alpha + \ell(m)}. \quad (12)$$

Checks and predictions at $r = 20$:

- $m = 1$ (E0’s `ctx`): $f \approx 0.4/44.4 = 0.009$, i.e. 0.90 CpG per 100 bp. E0 measured 0.94. The approximation is good because the TG/CA decay products are themselves one step from CG.
- Far flank in `P_meth` ($m = 0.953$): $\ell = 42.2\alpha$, giving 0.94 per 100 bp.
- Next to the bound motif ($m = 0.053$): $\ell = 8.0\alpha$, giving 4.8 per 100 bp, a $\sim 5\times$ **enrichment with zero selection on CpGs**. It falls to $\sim 1.3\times$ at $d = 25$, where $m = 0.70$.

This ignores dependence between neighbouring positions and the motif's own polymorphism. The oracle computes the exact neutral-flank expectation, and SLiM includes everything.

The feedback case. With $a_1 > 0$, retention lowers m (through ρ), which raises retention further. The map $f \mapsto f(m(\rho(f)))$ can have two stable fixed points: a methylated, CpG-poor state and an unmethylated, CpG-rich island. Whether an island self-sustains after the TF is lost (M0_meth_fb) or needs the TF to seed it (P_meth_fb) is an empirical question for E1.

10 P0: per-CpG priors across species

Extend the two-state chain of Section 9 with selection $S = 2Ns$ favouring the CpG (from selection on the CpG, or from gBGC). Under origin-fixation, the loss and gain rates become

$$a = \alpha \ell(m) \varphi(-S), \quad b = \alpha \gamma \varphi(S), \quad \varphi(S) = \frac{S}{1 - e^{-S}}, \quad (13)$$

where φ is $2N \times$ Kimura's u relative to neutral, and m is *germline* methylation. Because $\varphi(S)/\varphi(-S) = e^S$, the stationary CpG frequency is $f = \gamma e^S / (\gamma e^S + \ell(m))$: the neutral stationary distribution weighted by $e^{S \cdot \text{state}}$, as in Sella & Hirsh. For two species separated by total branch length T , measured through neutral non-CpG divergence $D = 3\alpha T$, reversibility gives

$$\mathbb{P}(\text{CpG in B} \mid \text{CpG in A}) = f + (1 - f) e^{-(a+b)T}. \quad (14)$$

Polymorphism. For new mutations with $\sigma = 2Ns$, the Poisson-random-field (Sawyer–Hartl) density of derived-allele frequency, per unit θ , is

$$\phi_\sigma(x) = \frac{1 - e^{-\sigma(1-x)}}{(1 - e^{-\sigma})x(1-x)} \xrightarrow{\sigma \rightarrow 0} \frac{1}{x}. \quad (15)$$

Expected segregating sites in a sample of n are $\theta \int_0^1 \phi_\sigma(x) [1 - x^n - (1 - x)^n] dx$; SLiM agrees at $\sigma = 0, -1, -2, -4, -8$. A change in germline m rescales polymorphism and divergence of CpG-loss alleles by the same factor $\ell(m)$, so P/D is unchanged. Selection gives $P/D \propto \int \phi_{-S}/\varphi(-S)$, which is 1.5, 2.4 and 7.8 at $S = 1, 2, 4$. This is the McDonald–Kreitman-type contrast that separates mutation from selection, where divergence and o/e alone lie on a ridge in (m, S) .

11 E2: stabilising selection on a sequence-encoded trait

Let a variant change one haplotype's readout by Δg , in a population whose mean E is near E^* . Then the heterozygote's log-fitness changes by

$$\log \frac{w_{\text{het}}}{w} = -\kappa \left[\left(E - E^* + \frac{\Delta g}{2} \right)^2 - (E - E^*)^2 \right] = -\kappa \Delta g (E - E^*) - \kappa \frac{\Delta g^2}{4}. \quad (16)$$

At the optimum the first-order term vanishes, so $s_{\text{het}} \approx \kappa(\Delta g/2)^2$: selection is *quadratic* in the effect. This has three consequences.

- Small-effect variants are nearly neutral. Under a degenerate readout ($R = \sum_i m_i$, one CpG at a time) each step is weak, but the sum is held tightly.
- With the trait additive, stabilising selection acts against the *minor* allele whatever the sign of its effect: an apparent underdominance of order $\kappa \Delta g^2$ (Robertson 1956). Effect–frequency tests therefore use $|\Delta g|$.

- Compensation. A loss followed by a gain elsewhere returns R to the optimum, so long-run substitution at individual CpGs is faster than a directional model with the same s predicts. P0's directional P/D is an upper bound.

Turnover between optimal genotypes needs every intermediate step to be weak. P's motif has 28 optimal configurations, but each one-step neighbour costs $2Ns \approx 14\text{--}72$, so turnover is negligible. C's steps cost $2Ns \approx 2\text{--}11$. At $\theta = 0.024$ CpG positions turn over almost completely while R is conserved. At realistic $\theta = 0.001$ (E2b) this holds only for weak steps ($2Ns \approx 1.6$). With $2Ns \approx 8\text{--}11$ positions mostly freeze: compensation then requires the deleterious intermediate (a lost CpG) to fix before a compensating gain arises. At high θ the gain can arise while the loss is still segregating (stochastic tunnelling), so the turnover rate of a degenerate trait depends on θ as well as on $2Ns$.

12 E3: combined and methylation-sensitive readouts

CP: $g = O e^{-R/R_0}$. The optimum is a curve $O = E^* e^{R/R_0}$ in the (R, O) plane, so stronger binding can be offset by more methylation. Because O takes discrete values (by match count), the curve is met at a few isolated points: (6 matches, $R \approx 2.5$) and (7 matches, $R \approx 4.2$). These are alternative stable states separated by valleys. **MS:** $g = O(1 - m_c)$, where m_c is the methylation of the motif CpG, which is required for binding. With protection $m_c = \sigma(a_0 - a_2 O)$, a bound site self-protects ($m_c \approx 0.09$ at O_7), so $\partial g / \partial m_c$ is rarely exercised. MS and the passenger control then have the same genotype–fitness map up to a constant.

13 Assumptions made explicit

1. Methylation is regenerated from sequence each generation. There are no epialleles.
2. Methylation is a cell-averaged deterministic value. Stochastic cell-to-cell noise would matter only through nonlinearity in fitness.
3. Cis only. There are no trans factors, so the TF's concentration is constant.
4. There is no methylation $\rightarrow$ TF arrow (methyl-sensitive TFs such as NRF1 are deferred). This is what makes P a “pure” passenger.
5. No recombination and no gBGC within the locus.
6. Constant N , a single population, and no demography. Two-population splits come in E3.
7. Codominant, additive expression. Fitness is Gaussian in E with the optimum at the maximum.

14 Parameters (E1 baseline, $Q = 2$)

symbol	value	meaning
N	100	diploids ($Q = 1$: 200)
L	300	bp
α	2×10^{-5}	per target per generation; $u = 3\alpha$, $\theta = 4Nu = 0.024$
r	20 (5, 50)	CpG transition multiplier
β, S_0	2, 6	occupancy curve
a_0, a_1, a_2	3, 0 or 0.5, 6	methylation map
λ, W	25 (10, 50), 25	protection range, density window (bp)
κ	11.7	so that $2Ns_{8/7} = 10$
burn-in, measurement	$5/u$, $2/u$ generations	composition equilibrates on the $1/u$ timescale, not N

The burn-in is set by $1/u$ rather than by N . Base composition (CpG content) changes only by substitution, which takes $\sim 1/u$ generations per site regardless of N . A $10N$ burn-in equilibrates diversity but not CpG content.